

JEE MAIN REVISION MASTER: STRAIGHT LINES AND PAIR OF STRAIGHT LINES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Forms of Straight Line & Distance Formulas	8 PYQs	Coordinate Geometry
Moderate	Pair of Straight Lines Through Origin	4 PYQs	Pair of Lines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Forms of Straight Line & Distance Formulas	$m_1 m_2 = -1$ iff $L_1 \perp L_2$, $d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$, $d_{\text{parallel}} = \frac{ c_1 - c_2 }{\sqrt{a^2 + b^2}}$
Pair of Straight Lines Through Origin	$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a+b}$, $a + b = 0$ iff Lines are perpendicular, $h^2 = ab$ iff Lines are coincident

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Forms of Straight Line & Distance Formulas (High Yield — 8 PYQs)

Core Principle: Line equations: Slope-intercept $y = mx + c$; Intercept form $(x/a) + (y/b) = 1$. Angle between lines: $\tan \theta = |(m_1 - m_2) / (1 + m_1 m_2)|$. Lines are parallel if $m_1 = m_2$; perpendicular if $m_1 m_2 = -1$. Perpendicular distance from (x_1, y_1) to $ax+by+c=0$: $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$. Distance between parallel lines: $d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$.

■ **Caution / JEE Pitfall:** Before using the parallel lines distance formula $d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$, ensure that the coefficients of x and y are made IDENTICAL in both equations!

■ **Speed Trick:** Line perpendicular to $ax + by + c = 0$ is simply $bx - ay + k = 0$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The line L_2 passing through $(3, -1)$ divides the line segment L_1 joining the points $(-1, 2)$ and $(3, 6)$ in the ratio 1:3 intern..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The system of equations $4x+6y=5$ and $8x+12y=10$ has..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A line passes through $(-1, -3)$ and perpendicular to $x+6y=5$. Its x intercept is..." [Ans: N/A]

3.2 Pair of Straight Lines Through Origin (Moderate Yield — 4 PYQs)

Core Principle: Homogeneous equation of second degree $ax^2 + 2hxy + by^2 = 0$ represents two straight lines passing through origin. Angle between them: $\tan\theta = \frac{\sqrt{h^2 - ab}}{a+b}$. Lines are perpendicular if $a + b = 0$ (coefficient of $x^2 +$ coefficient of $y^2 = 0$). Lines are coincident (parallel) if $h^2 = ab$.

■ **Caution / JEE Pitfall:** If $h^2 < ab$, the equation represents imaginary lines intersecting at $(0, 0)$.

■ **Speed Trick:** In KCET, if asked whether lines represented by $ax^2 + 2hxy + by^2 = 0$ are perpendicular, just check if $a + b = 0$ in 2 seconds!

Representative Source PYQs:

- **[2023 Shift] Q#1 (MCQ):** "A line passes through $(2,2)$ and is perpendicular to the line $3x+y=3$. Its y -intercept is..." [Ans: N/A]
- **[2022 Shift] Q#2 (MCQ):** "If the straight line $2x-3y+17=0$ is perpendicular to the line passing through the points $(7,17)$ and $(15,\beta)$, then β equals..." [Ans: N/A]
- **[2021 Shift] Q#4 (MCQ):** "The equation of straight line which passes through the point $(\cos^3\theta, \sin^3\theta)$ and perpendicular to $x\sec\theta + y\csc\theta = 1$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The line L_2 passing through $(3,-1)$ divides the line segment L_1 joining the points $(-1,2)$ and $(3,6)$ in the ratio 1:3 internally. The equation of line L_2 is:

- (A) $4x-3y-9=0$
- (B) $4x-3y+9=0$
- (C) $4x+3y-9=0$
- (D) $4x+3y+9=0$

Correct Answer: None

Solution Analysis: Let the points on line segment L_1 be $A(-1,2)$ and $B(3,6)$. The line L_2 divides this segment internally in the ratio 1:3. Using the section formula, the point of division P is $P\left(\frac{1 \cdot 3 + 3 \cdot (-1)}{1+3}, \frac{1 \cdot 6 + 3 \cdot 2}{1+3}\right) = P(0,3)$. So, line L_2 passes through the points $(3,-1)$ and $(0,3)$. Now find its slope: $m = \frac{3 - (-1)}{0 - 3} = \frac{4}{-3} = -\frac{4}{3}$. Using point-slope form with point $(0,3)$: $y - 3 = -\frac{4}{3}(x - 0)$ $3y - 9 = -4x$ $4x + 3y - 9 = 0$. So, the equation of line L_2 is $4x + 3y - 9 = 0$. Correct option

Benchmark Problem 2: KCET 2023 (Q#1)

A line passes through $(2,2)$ and is perpendicular to the line $3x+y=3$. Its y -intercept is

- (A) $\frac{2}{3}$
- (B) 1
- (C) $\frac{4}{3}$
- (D) $\frac{1}{3}$

Correct Answer: None

Solution Analysis: Equation of the line passing through $(2,2)$ and perpendicular to $3x+y=3$ is $(x-2)-3(y-2)=0 \Rightarrow x-3y=-4 \therefore y$ intercept of this line is $\frac{4}{3}$.